

Chapter 1

Monetary Conditions and FDI in ASEAN-Oriented Economies

1.1 Purpose and Scope

This report summarizes the current empirical specification and results for the relationship between monetary conditions and FDI net inflows in an unbalanced panel of ASEAN-oriented economies over 2000–2023. It is written as a clean replacement for the older methodology draft and is aligned with the latest regression tables.

1.2 Empirical Design

The empirical analysis estimates both static and dynamic specifications to examine the relationship between monetary conditions and FDI net inflows. The dependent variable is FDI net inflows as a share of GDP.

The static panel models are specified in the following general form:

$$FDI_{i,t} = \alpha + \mathbf{M}'_{i,t}\beta + \mathbf{C}'_{i,t}\gamma + \mu_i + \lambda_t + \varepsilon_{i,t} \quad (1.1)$$

where $FDI_{i,t}$ represents FDI net inflows as a share of GDP for country i in year t . The vector $\mathbf{M}_{i,t}$ contains the monetary variables of interest (broad money growth and, depending on the specification, the real interest rate). The vector $\mathbf{C}_{i,t}$ represents the set of control variables, which includes macroeconomic fundamentals (GDP growth, log GDP per capita, trade openness, exchange-rate depreciation, inflation) and, in certain specifications, institutional quality (Regulatory Quality). The terms μ_i and λ_t denote country and year fixed effects, respectively, while $\varepsilon_{i,t}$ is the idiosyncratic error term.

To account for the persistence of FDI and potential endogeneity, we also estimate a dynamic model using the two-step System GMM estimator:

$$FDI_{i,t} = \alpha + \delta FDI_{i,t-1} + \mathbf{M}'_{i,t}\beta + \mathbf{C}'_{i,t}\gamma + \mu_i + \lambda_t + \varepsilon_{i,t} \quad (1.2)$$

where $FDI_{i,t-1}$ is the lagged dependent variable.

The specification sequence is:

- **Spec 1:** baseline controls only (omitting *RIR* and Regulatory Quality).
- **Spec 2:** baseline controls plus the real interest rate.
- **Spec 3:** baseline controls plus Regulatory Quality.
- **Spec 4:** baseline controls plus the real interest rate and Regulatory Quality (Equation 1.1).
- **Spec 5:** the full model plus lagged FDI, estimated with two-step System GMM (Equation 1.2).

The static models report country-clustered standard errors and include country and year effects; Spec 1 is estimated with random effects, while Specs 2–4 use fixed effects. The dynamic model uses year effects and Windmeijer-corrected standard errors.

1.3 Pre-Estimation Diagnostics

The diagnostic sequence follows the protocol in §9 of the study instructions: unit-root tests, multicollinearity checks, heteroskedasticity, serial correlation, and cross-sectional dependence. The results jointly motivate the use of two-way fixed effects with Driscoll–Kraay standard errors.

1.3.1 Panel Unit Root Tests

Table 1.1 reports Levin-Lin-Chu (LLC) and Im-Pesaran-Shin (IPS) tests for all model variables. Both tests use a constant-only specification and AIC-selected lags (maximum 1). The null hypothesis is a unit root (non-stationary series).

Table 1.1: Panel Unit Root Tests (LLC and IPS)

Variable	Levin-Lin-Chu			Im-Pesaran-Shin		
	Stat	p	Decision	$\bar{t}$	p	Decision
FDI (% GDP)	−10.45	< 0.001	I(0)	−3.06	< 0.001	I(0)
Broad money growth	−8.16	< 0.001	I(0)	−4.12	< 0.001	I(0)
Real interest rate	−10.69	< 0.001	I(0)	−3.81	< 0.001	I(0)
GDP growth	−11.19	< 0.001	I(0)	−3.58	< 0.001	I(0)
ln(GDP per capita)	−4.90	< 0.001	I(0)	−1.76	0.169	Mixed
Trade (% GDP)	−4.59	< 0.001	I(0)	−1.75	0.193	Mixed
Δ ln(Exchange rate)	−9.08	< 0.001	I(0)	−3.50	< 0.001	I(0)
Inflation	−12.45	< 0.001	I(0)	−3.84	< 0.001	I(0)
Regulatory Quality	−4.38	< 0.001	I(0)	−2.22	0.003	I(0)
Rule of Law	−6.64	< 0.001	I(0)	−2.57	< 0.001	I(0)

Note: All tests use a constant-only specification with AIC-selected lags (max 1). LLC null: unit root for all panels. IPS null: unit root for all panels. $p < 0.05$ rejects the null (stationary). ln(GDP per capita) and Trade show mixed evidence under IPS; all other variables are stationary under both tests. Estimation in levels is appropriate for the bulk of the series.

1.3.2 Multicollinearity (VIF)

Table 1.2 shows VIF values for the full specification (Spec 4). All baseline controls have acceptable VIFs (< 5). Regulatory Quality (VIF = 13.0) exceeds the problematic threshold of 10 when combined with $\ln(\text{GDP per capita})$ (VIF = 7.7), indicating collinearity between the institutional and income variables. This is the standard cross-country pattern and is the primary reason Specs 3 and 4 have wider standard errors for those regressors. VIFs for Specs 1 and 2 (without RQ) are all below 3.

Table 1.2: Variance Inflation Factors — Full Specification (Spec 4)

Variable	VIF	Flag
Broad money growth	2.11	Acceptable
GDP growth	1.54	Acceptable
$\ln(\text{GDP per capita})$	7.74	Moderate concern
Trade (% GDP)	3.29	Acceptable
$\Delta \ln(\text{Exchange rate})$	1.13	Acceptable
Inflation	2.69	Acceptable
Real interest rate	2.66	Acceptable
Regulatory Quality	13.04	Problematic

Note: VIF < 5 = acceptable; 5–10 = moderate concern; > 10 = problematic. Collinearity between RQ and $\ln(\text{GDP per capita})$ is a known limitation of within-country institutional estimates and does not affect Specs 1–2.

1.3.3 Heteroskedasticity, Serial Correlation, and Cross-Sectional Dependence

Three further tests are reported for the full specification:

- **Modified Wald test** (group-wise heteroskedasticity): $W = 421.1$, $p < 0.001$, $df = 7$. The null of homoskedastic group variances is strongly rejected, confirming panel-level heteroskedasticity.
- **Wooldridge (2002) test** (first-order serial correlation): $t = 3.47$, $p = 0.010$, $df = 7$. The null of no AR(1) serial correlation is rejected, indicating that within-entity errors are autocorrelated.
- **Pesaran (2004) CD test** (cross-sectional dependence): $CD = 0.58$, $p = 0.560$, $N = 8$ countries, 28 pairs. The null of no cross-sectional dependence is not rejected.

Implication. Heteroskedasticity and serial correlation are both present; cross-sectional dependence is not. The aggregate Pesaran CD statistic ($CD = 0.58$, $p = 0.560$) already reported above confirms that error cross-sectional independence is a reasonable baseline assumption. Driscoll–Kraay standard errors, which are robust to both heteroskedasticity and serial correlation, are therefore the appropriate choice for the static models.

Table 1.3: FDI Determinants: Static Panel Estimates

Variable	Spec 1	Spec 2	Spec 3	Spec 4
Broad money growth	0.060*** (0.022)	0.109*** (0.027)	0.064** (0.029)	0.108*** (0.028)
Real interest rate		0.099** (0.041)		0.104*** (0.033)
GDP Growth	0.112 (0.084)	0.405*** (0.112)	0.312** (0.143)	0.410*** (0.130)
ln(GDP per capita)	0.877*** (0.175)	2.235*** (0.676)	2.997** (1.173)	1.715* (1.005)
Trade openness (% GDP)	-0.009 (0.012)	-0.025 (0.021)	-0.026 (0.022)	-0.024 (0.023)
$\Delta \ln(\text{Exchange rate})$	-0.009 (0.027)	-0.000 (0.040)	0.019 (0.033)	-0.002 (0.033)
Inflation	0.017 (0.038)	0.151* (0.078)	0.061 (0.070)	0.160*** (0.060)
Regulatory quality			-2.032 (1.821)	1.076 (2.218)
N	216	192	207	184
Within R^2	0.192	0.257	0.252	0.243
Estimator	RE	FE	FE	FE

Note: Standard errors are in parentheses. *p<0.10, **p<0.05, ***p<0.01.

1.4 Results

1.4.1 Static panel estimates

The clearest result is the broad money coefficient. It is positive and statistically significant in all four static specifications, providing the strongest evidence in favour of a liquidity channel. The real interest rate does not deliver a stable negative effect — it is positive and significant in Specs 2 and 4 — so the data do not support a simple cost-of-capital story. Among the controls, GDP growth and log GDP per capita are generally positive and significant; trade openness, exchange-rate depreciation, and inflation are mostly imprecise. Regulatory Quality is not robust across static specifications.

1.4.2 Dynamic GMM estimate

The dynamic GMM estimate is mixed.

Lagged FDI is negative and significant, Regulatory Quality is positive and significant, and trade openness is positive and significant. However, the AR(2) test rejects the null of no second-order serial correlation ($p = 0.0002$), so this specification should be treated cautiously. The model uses 64 instruments; the Hansen test p -value is 1.0 (instrument proliferation is a concern) and AR(1) $p = 0.4484$.

Table 1.4: FDI Determinants: Dynamic GMM Estimate

Variable	Spec 5
Lagged FDI (% GDP)	-2.432** (1.174)
Broad money growth	-0.036 (0.675)
Real interest rate	0.545 (0.979)
GDP Growth	0.062 (0.132)
ln(GDP per capita)	-3.011** (1.176)
Trade openness (% GDP)	0.276*** (0.081)
$\Delta \ln(\text{Exchange rate})$	-0.020 (0.142)
Inflation	0.592 (0.942)
Regulatory quality	0.672*** (0.225)
N	320
Estimator	2-step GMM

Note: Standard errors are in parentheses. *p<0.10, **p<0.05, ***p<0.01.

1.5 Interpretation

The most defensible reading is that liquidity conditions are more consistently associated with FDI inflows than interest-rate conditions in this panel. The static models provide the cleanest pattern. The dynamic model adds a persistence result, but its diagnostic weakness makes it less persuasive as a headline finding.

1.6 Caveats

- Sample sizes differ across specifications, so coefficients are not directly comparable across every model.
- The dynamic model has a concerning AR(2) diagnostic.
- The static models explain only a modest share of within-country variation.

1.7 Bottom Line

Broad money growth is the most robust correlate of FDI inflows in the current results. Interest-rate and institutional effects are less stable and more model-dependent, so the report should stay moderate in its claims.